Role *A* Role *B*

$\mathsf{a_enc}(\langle A, N_a\rangle, \mathsf{pk}(B))$

A initiates and issues a challenge

$\mathsf{a_enc}(\langle N_a, \langle N_b, B\rangle\rangle, \mathsf{pk}(A))$

B responds and issues their own challenge

$\mathsf{a_enc}(N_b, \mathsf{pk}(B))$

A responds to *B*'s challenge